

The Wave Machine Lab

Grade Level: 9th–12th Grade

Subject: Science

Duration: 1 Hr.

Overview:

This lab will demonstrate how to build a wave machine that will simulate a transverse wave. Participants will be observing wave properties such as amplitude, frequency and wavelength, speed, and velocity. Understanding the properties of waves and how they move through mediums can help scientists understand materials that they cannot see with their eyes. Specifically, geophysicists use seismic waves to understand the different material layers that sit below our feet. Analyzing the seismic wave properties such as amplitude, frequencies, wavelength, speed, and velocity is how they are able to determine where earthquakes occur, identify underground resources and hazards, and determine the structure of the earth.

For this lab, the seismic wave that is related to this model is a s-wave, shear wave. The s-wave is a transverse wave that moves through the body of the earth. This wave moves through solids, and its velocity increases with depth and pressure. However, it cannot move through liquids as liquids lack shearing motion. In the beginning of this lab, we will simulate a one-layer model, meaning the same material. Then we will transition to a two-layered model; two different materials are present.

Objectives

By end of the lesson students will be able to:

- Identify and describe the characteristics of a wave.
 - Demonstrate an understanding of how waves move through materials.
 - Connect physics properties to earth processes.
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Materials

- Duct tape
 - Wooden Skewers (~ 30 cm or 12 inches in length)
 - Gummy Bears (big bag)
 - 2 Anchoring Items
 - Timer (stopwatch or phone)
 - Ruler/ meter stick
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Set up: 5-15 minutes

Have students break into groups of 4-5 and send two to collect materials, while the other two begin reading the following steps.

1. Set Anchoring items (chairs, or support stands) parallel from one another 1.5 m to 3 m apart (~ 5-10 ft)
2. Attach Duct tape to one anchor, sticky side facing up, and then have someone hold the anchor and pull the tape to the other anchor and attach again with sticky side facing upward. The duct tape needs to be held in tension between the two anchors, not a lot of drooping! The anchors should be something ridged like clamp stands or desk chairs.
3. Mark the center of the skewers and place them on the duct tape ~ 3cm (1 in) apart along the tape till you reach the next anchor.
4. Optional: Place another strip of duct tape over the skewers to sandwich them in, to create a more “robust” wave machine.
5. Place 1 Gummy bear on each end of the skewers; some of the skewers may become lopsided. Adjust the gummy bears along the length of the skewers until they are leveled.

Experiment: 30 – 40 minutes

1. Begin by picking a starting point along the wave machine, preferably one of the skewers near the end of the machine. This will be the skewer you initially lift and drop to create the source of the wave.
2. Measure the distance from your source to the end of the machine, have 1 person stand at that end with a stopwatch in hand. The person at the end will be representing the receiver; they will start their timers the moment the energy is released from the initial starting point and end the timer when the wave reaches them at the end.
3. The person at the starting point should lift one end of the skewer ~ 12cm (5 in) and then let it drop. The height you lift and drop the end of the skewer from will be the amplitude of the wave.
4. You will observe the wave propagate along the wave machine until it reaches the end and then reflect towards the starting point. Make sure to take your time measurement when the wave reaches the end of the machine and not when it reflects back.
5. With the distance and time measurements collected you will now solve for the wave speed: $S = \text{Distance} / \text{Time}$, and velocity since we know the wave direction: $V = \text{Displacement} / \text{Time}$.
6. Using the same starting and end points repeat 3-5, with varying amplitudes. (Change the height from where you drop the skewer). Does speed or velocity change?

Amplitude (cm)	Distance (cm)	Time (sec)	Speed (cm/sec)	Velocity (cm/sec)
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7. Next, lift the skewer and instead of letting it drop, give it a little push down. Again, time the wave and calculate the speed and velocity. You will observe the wavelength decrease and the wave frequency increases. Does the wave speed increase?

Distance (cm)	Time (sec)	Speed (cm/sec)	Velocity (cm/sec)

8. Next, remove the gummy bears from a little over half of the skewers, leaving ~ 10 – 20 skewers at one end of the wave machine with gummy bears. This will represent a change in material, due to mass and density changes.
9. Have the person at the starting point, preferably on the end of the machine with the gummy bears, initiate the wave.
10. Observe how the wave moves through the 2 different “materials”. What do you see? Write a description or draw a picture!
11. Measure the distance from the starting point to the boundary, the last skewer with gummy bears on it, and then measure the distance from the boundary point to the end of the machine.
12. Have the person with the stopwatch use the lap function to mark where the wave passes the boundary point and then end the time when it reaches the end of the machine.
13. With the time and distance values solve for the velocity of the wave. How does the velocity of the wave change as it goes between materials?

Material	Start Distance- x1 (cm)	End Distance- x2 (cm)	Time (sec)	Velocity (x2-x1)/T
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1				
2				